Effects of Sound Waves

Exam code: 0625 & 0972

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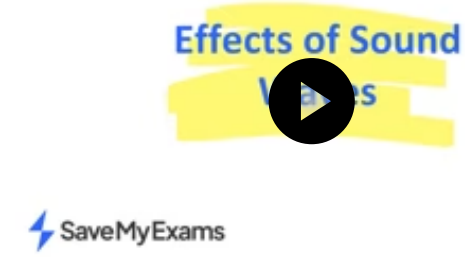

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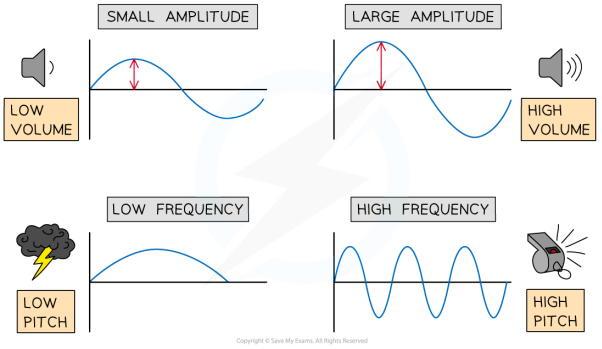
Help me understand

Pitch & loudness

- Changes in **amplitude** and **frequency** affect the **loudness** and **pitch** of sound waves
- The **frequency** of a sound wave is related to its **pitch**
  - Sounds with a **high** pitch have a **high** frequency (or short wavelength)
  - Sounds with a **low** pitch have a **low** frequency (or long wavelength)
- The **amplitude** of a sound wave is related to its **volume**
  - Sounds with a **large** amplitude have a **high** volume
  - Sounds with a **small** amplitude have a **low** volume



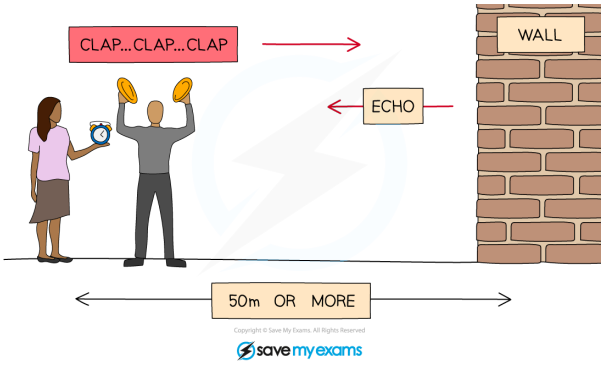
Pitch and amplitude of sound

Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it

Echoes

- Sound waves **reflect** off hard surfaces
  - The reflection of a sound wave is called an **echo**

An example of an echo



A clap sound is reflected off a wall. When the reflected sound is heard, this is the echo

- The distance to the surface where the echo reflects can be calculated using the equation:

$$\text{distance} = \frac{1}{2} \times \text{speed} \times \text{time}$$

- Where:
  - distance = half the total distance travelled by the sound wave, measured in metres (m)
  - speed = speed of sound in air, measured in metres per second (330 - 350 m/s)
  - time = total time for sound wave to travel to the surface and back, measured in seconds (s)

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